

Problem B. Fixed Tour

Input file: standard input
Output file: standard output

You are given a tree consisting of n vertices, numbered from 1 to n . A **tree** is a connected undirected graph with $n - 1$ edges and no cycles.

We perform a **Depth First Search** (DFS) starting from a chosen root vertex. During the DFS, we maintain a global **timer**, initially set to 0. When the DFS visits a vertex v :

1. Increment the *timer* by 1.
2. Set the entry time $tin_v = timer$.
3. For each neighbor u of v that has not been visited yet, recursively call the DFS for u .

Crucially, the neighbors of any vertex are processed in the order they appear in the input data.

A vertex v is called **lucky** if its entry time is equal to its index: $tin_v = v$.

Your task is to calculate the number of lucky vertices for every possible choice of the root. Specifically, for each i from 1 to n , find the number of lucky vertices if the DFS starts at vertex i .

Input

The first line contains an integer n ($1 \leq n \leq 10^5$) — the number of vertices.

Each of the next $n - 1$ lines contains two integers u_i and v_i ($1 \leq u_i, v_i \leq n, u_i \neq v_i$), representing an edge between u_i and v_i .

Output

Print n integers in a single line. The i -th integer should be the number of lucky vertices if vertex i is the root of the tree.

Scoring

This problem contains 6 subtasks.

The degree of a vertex is the number of edges that are incident to the vertex.

Subtask	Additional Constraints	Points
0	Examples	0
1	$n \leq 3000$	9
2	Each vertex has degree at most 2	18
3	Only one vertex has degree more than 1	18
4	For each $i \in [1, n]$, an edge exists to vertex $2i$ if $2i \leq n$, and to $2i + 1$ if $2i + 1 \leq n$.	14
5	$u_i = i$ and $v_i > i$ for all $1 \leq i < n$	19
6	—	22

Examples

standard input	standard output
3 1 2 1 3	3 1 0
10 6 2 4 9 3 8 2 3 8 10 5 10 5 9 6 7 1 4	1 0 0 0 2 2 0 0 0 1